Fiscal Year:	FY 2019	**Task Last Updated:**	FY 12/28/2019
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Mitigating Radiation-induced Bone Loss via Dietary Modulation of Inflammatory Cytokines		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	12/07/2016	**End Date:**	08/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/31/2019 per NSSC information (Ed., 1/7/19) NOTE: Extended to 12/6/2018 per NSSC information (Ed., 3/20/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hogan, Harry Ph.D. (Texas A&M University) Turner, Nancy Ph.D. (Michigan State University (formerly at Texas A&M University AgriLife Research, until 2018))		
Grant/Contract No.:	NNX17AC91G		
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Task Description:	Proposal Summary: Astronauts traveling to Mars or other planetary surfaces will be exposed to low doses (up to 2 Gy) of high-energy radiation particles that compose galactic cosmic radiation (GCR). A growing body of literature documents a rapid acceleration of bone resorption activity and some suppression of bone-forming cells after exposure to even very low doses of space-relevant radiation. One of the key mechanisms proposed for these changes is an increase in pro-inflammatory cytokines like TNF-a. We already know that International Space Station (ISS) crew members consuming a diet rich in omega-3 Fatty acids (FAs) experience smaller reductions in bone mineral density over 6-month missions than do fellow astronauts consuming a diet low in these FAs. This project proposes to investigate the impact of consuming a diet high in omega-3 FAs on radiation-induced pro-inflammatory cytokines. Hence, this simple dietary intervention may provide a low-cost, low-risk means of countering the harmful effects of radiation on bone integrity. A currently funded NASA Space Biology study based at Texas A&M (Principal Investigator (PI): Dr. Nancy Turner) will expose multiple cohorts of mice at Brookhaven National Laboratory's NASA Space Radiation Laboratory (NSRL) facility to space relevant doses of high energy iron particles) over the next 12 months. Her project will determine the selective impact of radiation on radio-sensitive intestinal stem cells in two sets of animals: those consuming a corn oil-supplemented diet simulating the average American diet and those consuming a diet supplemented with fish oil, which elevates dramatically omega-3 FA intake. The PI's current collaborations with Dr. Turner and our laboratories' close proximity have enabled an unparalleled tissue sharing opportunity. We propose to collect both long bones and serum from these animals to test our over-arching hypothesis: a diet high in omega-3 FAs will mitigate radiation-induced bone loss by reducing the generation of inflammatory cytokines in bone tissue. The study offers the possibility of comparing responses to 3 different space-relevant doses (0.1, 0.25, and 0.5 Gy) of 56Fe at 1 Gev/nucleon, along with a gamma reference group (0.25 to 2 Gy). Analyses of specimens collected at 12 hours post-exposure will focus on changes in pro-inflammatory cytokines in serum and in bone cells called osteocytes, which in turn signal to both bone-forming and bone-resorbing cells. The two later post-irradiation time points proposed (4 wk and 8 wk) will enable us to determine early and late effects on bone cell activity and bone structural integrity, important since the current data base on time course of bone alterations post-exposure is sparse. Significance : Given that the expensive NSRL (NASA Space Radiation Laboratory) live animal experiments are already funded, this project can yield a wealth of new data important to minimizing fracture risk for exploration class missions at minimal extra cost to NASA. With this project we will generate a comprehensive assessment of the impact of 3 doses of high-energy iron particles on bone integrity, at 3 dose levels and at 3 time points post-irradiation. Comparisons with gamma-irradiated groups at comparable doses will allow for RBE (relative biological effectiveness) determinations; sham irradiated animals will be sacrificed at the same time points. After just one year, we will have proof-of-concept and feasibility testing completed for a high omega-3 FA diet as a countermeasure to radiation-induced bone loss, and interesting science data on the role of osteocytes in the pro-inflammatory response to radiation. Should it prove successful, this countermeasure is ready for immediate operational implementation, as omega-3 fatty acid-rich diets and/or supplements are available now to ISS crew. These data will also be relevant to clinical patients undergoing radiotherapy and patients with chronic inflammatory disease (e.g., inflammatory bowel disease), all of whom experience significantly elevated fracture rates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With this project we will generate a comprehensive assessment of the impact of 3 doses of high-energy iron particles on bone integrity, at 3 dose levels and at 3 time points post-irradiation. Comparisons with gamma-irradiated groups at comparable doses will allow for RBE (relative biological effectiveness) determinations; sham irradiated animals will be sacrificed at the same time points. After just one year, we will have proof-of-concept and feasibility testing completed for a high omega-3 FA diet as a countermeasure to radiation-induced bone loss, and interesting science data on the role of osteocytes in the pro-inflammatory response to radiation. Should it prove successful, this countermeasure is ready for immediate operational implementation, as omega-3 fatty acid-rich diets and/or supplements are available now to ISS crew. These data will also be relevant to clinical patients undergoing radiotherapy and patients with chronic inflammatory disease (e.g., inflammatory bowel disease), all of whom experience significantly elevated fracture rates.
	Experimental Approach and Methods: Bones and serum collected from animals exposed to 56Fe and gamma radiation by the parent protocol at multiple doses (56Fe: 0.1, 0.25, and 0.5 Gy of 1000 MeV/n , 25 cGy/min at Brookhaven National Lab; gamma X-ray radiation: 0.2, 0.5, and 2 Gy, at Texas A&M) and time points (12 hr, 4 wk, 8 wk post-exposure) were stored by appropriate procedures to enable the proposed outcomes detailed below. One half of all animals were fed a corn oil-cellulose (COC) and the other half a fish oil-pectin (FOP) diet to test the impact of an "anti-oxidant" diet on inflammatory changes following irradiation. Tissues collected at 12 hours post-radiation exposure were assessed only for changes in serum TNF-a, serum TRAP5b (a marker for osteoclast number and therefore resorption activity) using commercial ELISA assay kits, and by immunohistochemical (IHC) staining for TNF-a and sclerostin (see Metzger et al., 2017 for methods) in metaphyseal bone osteocytes. Assessment of %-positive osteocytes for select proteins is, we believe, more useful than assessing altered gene expression of homogenized bone tissue, since the latter cannot confirm that increases in mRNA result in more protein product, nor is it localized to any one cell type. Any changes in bone structure would be too small to detect at this early time point. Tissues collected at 4 and 8 weeks post-radiation exposure were assessed for changes in serum TNF-a and IHC staining for osteocyte TNF-a and sclerostin [in R distal femur] and, in addition, by 1) micro-CT (in consultant Dr. Larry Suva's laboratory, TAMU Dept. of Veterinary Physiology & Pharmacology) for distal femur bone structural properties; 2) histomorphometry of distal femur for evidence of bone formation/ resorption activity, using an epifluorescent microscope interfaced with a CCD camera and OsteoMeasure software; and 3) [not yet completed] mechanical testing of mid-shaft tibiae and L femoral neck, using a desktop Instron device. Earlier annual reports included data on male mice; since very few significant effects were observed in those males, we proceeded to focus only on female mouse data. Hence, all results discussed below are for female mice only Statistical analyses were performed on all outcome measures using a two-way MANOVA (factors = diet, dose) with each time point and ion species analyzed separately and an alpha level of 0.10 set a priori. We believe using this higher p value threshold is appropriate for this project, given the very low n's in some groups and the exploratory nature of these experiments, erring on the side of inclusiveness. Results: (organized by Specific Aims) Specific Aim 1: Determine alterations in bone structural integrity (density, geometry, microarchitecture, mechanical

properties); bone cell activity; serum TNF-a, and osteocyte TNF-a and sclerostin, in mice exposed to low dose HZE (high energy particle) radiation.

• Hypothesis 1: An early up-regulation of serum TNF-a and TNF-a in osteocytes, after radiation exposure will be associated with increased resorption and decrements in bone structural integrity observed 4 and 8 weeks later; increased osteocyte sclerostin by 4 weeks will be associated with declines in bone formation activity.

Little of our original working hypothesis about radiation exposure effects in our control (COC-fed) mice is borne out by our results. We do not observe any significant decrements in bone structural integrity, except for mice at 8 weeks after gamma exposure, when some significant declines were observed in distal femur cancellous bone mass indices (%BV/TV and vBMD) and in trabecular connectivity. The only group to exhibit any change in our bone formation index (% osteoid surface/total surface) were 56Fe–exposed mice after 8 weeks; this suggests that if we had tracked these mice for a longer time, we might ultimately observe a corresponding decline in cancellous %BV/TV. We found no evidence for an early pro-inflammatory response as measured by serum and osteocyte TNF-a (%+Ot.TNF-a), but rather a very late decrease in serum TNF-a 8 weeks after gamma exposure. Sclerostin, a negative regulator of bone formation, declines a small amount in cortical bone osteocytes 4 weeks after 56Fe exposure; IGF-I, a positive regulator of bone formation, appears to be slightly upregulated by 56Fe exposure in both cortical and cancellous bone compartments. Specific Aim 2: Determine impact of a diet high in omega-3 fatty acid content on osteocyte TNF-a and decrements in bone structural integrity after exposure to low dose HZE radiation.

• Hypothesis 2: Mice consuming a high omega-3 FA diet will exhibit reduced serum TNF-a; reduced sclerostin and TNF-a in osteocytes; and attenuated decrements in bone structural integrity after radiation exposure.

Task Progress:

Cited below are relevant results when there was a significant main effect of diet and, in particular, a diet by dose interaction.

We originally hypothesized that consumption of a diet high in omega-3 fatty acids (FOP) would result in a diminished increase in serum and osteocyte TNF-a, and in osteocyte sclerostin, which would explain mitigated decrements in structural integrity, following radiation exposure. The dietary impact was most clear at the 4- and 8-week time points for serum TNF-a values; following 56Fe exposure, serum TNF-a was indeed lower in FOP-fed mice. Intriguingly, a different effect occurred following gamma exposure, when FOP-fed mice at 8 weeks exhibited on average much higher serum TNF-a values than did COC-fed mice. At both these time points weeks removed from the time of irradiation, increasing doses of radiation either had no effects or resulted in declines in serum TNF-a values. However, when examining %osteocytes positive for TNF-a, the FOP-fed mice exhibit the same or slightly higher %+Ot.TNF-a values in both gamma- and 56Fe-exposed mice. Hence, the presence of TNF-a in these important regulatory bone cells does not track well with serum TNF-a.

The most significant differences in bone structural integrity were observed in mid-shaft cortical bone diameters and the connectivity density in the cancellous compartment of the distal femur. Cortical bone diameters (but not area, or thickness of the cortical shell) at 8 weeks following 56Fe exposures were generally larger at all 56Fe doses in FOP-fed mice. A significant effect of diet was also detected for connectivity density following gamma exposure; at the final time point examined (8 weeks post-exposure) FOP-fed mice exhibited a higher values for Conn.Dens with increasing gamma dose (vs. a decline in Conn.Dens at 4 and 8 weeks for the COC-fed mice). Given that higher values of Conn.Dens and larger cortical diameters make important contributions to cancellous and cortical bone compartment resistance to fracture, respectively, these findings suggests that diets high in omega-3 fatty acids can be beneficial in protecting bone health.

Another important finding of this tissue-sharing project is that high energy 56Fe radiation exposure does not result in consistent decrements in bone structural integrity, at least up to 8 weeks following exposure. Most microCT-derived variables either did not change with increasing dose or were actually improved after 56Fe exposure (trabecular number and, perhaps, cancellous bone mass [%BV/TV]). The increased prevalence of osteocytes positive for IGF-I (%+Ot.IGF-1) in both cortical and cancellous bone 8 weeks after 56Fe exposure may provide one mechanism contributing to those improvements. On the other hand, lower energy gamma exposure resulted in a decline in %BV/TV, connectivity density (Conn.Dens), and volumetric bone mineral density (vBMD) of the DF cancellous bone after 8 weeks. These findings highlight the critical need to evaluate space radiation’s impact on tissue integrity using the high energy ion species found in galactic cosmic radiation, and not relying on studies utilizing low-energy gamma radiation.

CONCLUSION

This study is one of the first, to our knowledge, to document changes in osteocyte proteins following radiation exposure. Using immunohistochemical (IHC) staining, we tracked alterations in proportions of osteocytes positive for TNF-a, sclerostin and (for 56Fe exposure only), IGF-I (a protein anabolic to bone). There was modest evidence for a positive impact of the FOP diet on cortical bone osteocytes positive for TNF-a after radiation exposure: %+Ot. TNF-a declined (12 hr and 4 wk after gamma exposure) or held steady in FOP-fed mice vs an increase in COC-fed mice. These changes were concurrent with a few advantages to bone structural integrity observed following gamma exposure (increased connectivity density in cancellous bone at the final time point in FOP-fed mice). We are working on a manuscript to be submitted to Bone or to npjMicrogravity describing these findings in a more comprehensive manner.

Given that a diet high in anti-oxidants like fish oil has been previously demonstrated to mitigate reductions in DEXA-measured BMD after ISS flights, these data provide preliminary evidence for a mechanism if and when radiation exposure is involved: reduction of circulating TNF-a, a pro-inflammatory cytokine, after 56Fe exposure simulating GCR. The role of TNF-a in bone tissue itself (reflected by IHC staining of osteocytes, the most ubiquitous bone cell type) is less clear and did not track with the serum TNF-a changes. Dietary countermeasures are relatively inexpensive, safe, and hold much promise to mitigate both oxidative damage and inflammatory changes that have been demonstrated with prolonged sojourns in microgravity and especially after radiation exposure.

Bibliography Type: Description: (Last Updated: 01/20/2026)

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Abstracts for Journals and Proceedings	Little SE, Allaway HCM, Bokhari RS, Seidel DV, Wahl KL, Turner ND, Ford JR, Suva L, Bloomfield SA. "Can acute galactic cosmic radiation-induced bone loss be mitigated by dietary modulation of inflammatory cytokines? " 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Conference program/abstracts distributed at meeting. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Little SE, Allaway HCM, Bokhari RS, Seidel DV, Wahl KL, Turner ND, Ford JR, Suva L, Bloomfield SA. "Can acute galactic cosmic radiation-induced bone loss be mitigated by dietary modulation of inflammatory cytokines?" American College of Sports Medicine, Orlando, FL, May 28-June 01, 2019. Medicine and Science in Sports and Exercise. 2019 Jun;51(6 Suppl):406. https://doi.org/10.1249/01.mss.0000561717.28075.cd , Jun-2019
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